

Is this a successful proof of Fermat's Last Theorem?

by John B. van der Linden

[0] $a^n + b^n = c^n$ where $a < b < c$ and a, b, c , and n are positive integers with $n \geq 2$
since $c > a$ and $c > b$, c can be written as $x+a$ or $y+b$ where x and y are positive integers

[1] thus $a^n + b^n = (x+a)^n = (y+b)^n$

now expand $(x+a)^n$ and $(y+b)^n$:

$(x+a)^n = x^n + a^n + \text{remaining terms (call their sum "f")}$

$(y+b)^n = y^n + b^n + \text{remaining terms (call their sum "g")}$

thus $a^n + b^n = x^n + a^n + f = y^n + b^n + g$

subtract from both sides

$b^n = x^n + f$

$a^n = y^n + g$

f and g are both positive integers, thus $b^n > x^n$ and $a^n > y^n$

take n th root of both sides

$b > x$ and $a > y$

write b as $x+s$ and a as $y+r$ where s and r are positive integers

now back to [1]

$a^n + b^n = (x+a)^n = (y+b)^n$

$a^n + b^n = (x+(y+r))^n = (y+(x+s))^n$

$a^n + b^n = (x+y+r)^n = (x+y+s)^n$

thus $r = s$; rename this integer Z .

[2] then $a^n + b^n = (x+y+Z)^n$

by definition, $a \neq b$ and since $c = x+a = y+b$, $x \neq y$

[3] $y+Z = a$ and $y < a$, thus **a must be at least 2** (if $y = Z = 1$)

$b > a$ so **b must be at least 3**

[4] thus $a + b \geq 5$

now back to [2]:

$a^n + b^n = (x+y+Z)^n$

recall that $b = x+Z$ and $a = y+Z$. thus $x = b-Z$ and $y = a-Z$. Substitute these values:

$a^n + b^n = ((b-Z)+(a-Z)+Z)^n$

[5] $a^n + b^n = (a + b - 2Z + Z)^n = (a + b - Z)^n$

Substituting the minimum values of a, b , and Z into [5] gives

[6] $2^n + 3^n = (2 + 3 - 1)^n = 4^n$

Divide both sides by 4^n

$$\left(\frac{1}{2}\right)^n + \left(\frac{3}{4}\right)^n = 1$$

Both terms on the left are less than 1 and get smaller as n increases, thus n must be small.

Substituting n = 1 gives $\left(\frac{1}{2}\right) + \left(\frac{3}{4}\right) = 1.25$ which is greater than 1

Substituting n = 2 gives $\left(\frac{1}{4}\right) + \left(\frac{9}{16}\right) = 0.8125$ which is less than 1

All larger values of n will give a value for [6] that is less than 0.8125, since both exponent terms are decreasing

Thus n must be between 1 and 2 if a = 2, b = 3, and Z = 1. But this violates our initial condition that n is a positive integer

[7] Thus a = 2, b = 3, and z = 1 cannot be correct

[8] The next lowest possible value for a is 3. Let a = 3, b = 4, and Z = 1. Then we have

$$3^n + 4^n = (3 + 4 - 1)^n = (2 * 3)^n$$

$$3^n + 4^n = (2^n)(3^n)$$

$$1 + \left(\frac{4}{3}\right)^n = 2^n$$

$$2^n - \left(\frac{4}{3}\right)^n = 1$$

For n = 1 we have $2 - \frac{4}{3} = \frac{2}{3} \neq 1$, so n cannot be 1

For n = 2 we have $2^2 - \left(\frac{4}{3}\right)^2 = 4 - \left(\frac{16}{9}\right) = \left(\frac{36}{9}\right) - \left(\frac{16}{9}\right) = \frac{20}{9} = 2.222\dots \neq 1$, so n cannot be 2

But we know n CAN be 2, since $3^2 + 4^2 = 5^2$

Thus the problem must be with the minimum value of Z, not those of a or b

[9] So **Z must be at least 2**

[10] and **a + b >= 7**

Now return to [5].

$$a^n + b^n \leq (a + b - Z)^n$$

Z must be at least 2.

$$a^n + b^n \leq (a + b - 2)^n$$

We know the minimum value of the left side is $3^2 + 4^2 = 5^2 = 25$.

[11] Thus $25 \leq a^n + b^n \leq (a + b - 2)^n$

Substitute k = a+b

$$25 \leq (k-b)^n + (k-a)^n \leq (k-2)^n$$

Since b > a, we know $(k-b)^n < (k-a)^n$. Thus we can write

$$25 \leq 2^*(k-b)^n \leq (k-b)^n + (k-a)^n \leq (k-2)^n$$

Or simply

$$[12] \quad 25 \leq 2^*(k-b)^n \leq (k-2)^n$$

Write the factor 2 (from the middle part) as $(2^{1/n})^n$

$$25 \leq (2^{1/n})^n * (k-b)^n \leq (k-2)^n$$

Take the nth root of both sides

$$25^{(1/n)} \leq 2^{(1-n)}(k-b) \leq k - 2$$

Divide by $(k-b)$

$$(25^{(1/n)})/(k-b) \leq 2^{(1-n)} \leq (k-2)/(k-b)$$

Back-substitute $a + b = k$

$$(25^{(1/n)})/a \leq 2^{(1-n)} \leq (a+b-2)/a$$

$$[13] \quad (25^{(1/n)})/a \leq 2^{(1-n)} \leq 1 + (b/a) - (2/a)$$

The largest possible value of $(25^{(1/n)})/a$ is at $a = 3$. So

$$(25^{(1/n)})/a \leq (25^{(1/n)})/3 \leq 2^{(1-n)} \leq 1 + (b/a) - (2/a)$$

Ignore the rightmost part.

$$[14] \quad (25^{(1/n)})/a \leq (25^{(1/n)})/3 \leq 2^{(1-n)}$$

Raise everything to the power of n .

$$[15] \quad 25 / (a^n) \leq 25 / (3^n) \leq 2$$

Ignore the leftmost part.

$$[16] \quad 25 / (3^n) \leq 2$$

$$3^n / 25 \leq 1/2$$

$$3^n \leq 25/2$$

$$n \leq \log_{\text{base } 3} \text{ of } 12.5$$

$$n \leq 2.2990\dots$$

$$[17] \quad \text{The only possible integer value of } n \text{ is } 2.$$

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Note: I was stuck at [6], and decided to ask Google's AI Overview tool, "do the graphs of $f(n) = (2^n + 3^n)$ and $g(n) = 4^n$ intersect anywhere?" It stated yes, at only one point ($n = 1.507\dots$) (I failed to specify that n needed to be an integer). It then provided the technique of dividing both sides by 4^n , noting the fact that both left-hand terms decreased, and evaluating the equation for $n = 1$ and $n = 2$.

The rest of the proof is my own original work.